

E Case Studies

We perform a small series of case studies to demonstrate the effectiveness of our proposed alignment techniques in practice. To make results relevant to real-world applications and create a challenging setup for our methods, we use a 6,619-question test set subset where the questions’ answers changed recently. More specifically, we only use questions from our test set with ground truth answers from 2020 to 2022 that share no common element to the answers in 2019, which is the knowledge peak of the LLaMa2-70B model as shown in Figure 1.

In detail, we compare the model predictions between the unaligned, target-year (2022) finetuning, and temporal-adaptive finetuning models. To simplify our analysis, we categorize the model predictions into three: 1) correct, where the model’s prediction hits one of the ground truth answers in 2022; 2) misaligned, where the model’s prediction hits one of the answers that are historically correct, but not in 2022; 3) incorrect, where the model’s prediction does not hit any of the answers in the question’s historical answer set. We present the number of answers predicted by the unaligned and target-year finetuning models categorized into the three classes above in Table 18.

		Unaligned		
		Correct	Misaligned	Incorrect
Target-Year Finetuned	Correct	186	522	307
	Misaligned	57	924	553
	Incorrect	89	765	3216
Total		332	2211	4076

Table 18: The quantity of correct, misaligned, and incorrect answers predicted by unaligned and target-year (2022) finetuned models.

E.1 What questions can the LM answer correctly without alignment, even if their answers changed recently?

From Table 18, we find that the unaligned LLaMa2-70B model can still answer 332 time-sensitive questions correctly. We show four examples from this subset in Table 19.

Question	Unaligned Model’s Answer	Answer Correct Years
Who is the current captain of Leicester City F.C.?	Kasper Schmeichel	2021, 2022
What is the official United Nations theme for International Women’s Day?	Gender equality today for a sustainable tomorrow	2022
What character does Jeri Ryan portray in her current television series?	Seven of Nine	2020, 2021, 2022
Who holds the position of offensive coordinator for the Pittsburgh Steelers at the moment?	Matt Canada	2021, 2022, 2023

Table 19: Examples of questions that can be answered correctly as of 2022 by the unaligned LLaMa2-70B model, even if their answers differ from those in 2019.

In addition, we want to check what kinds of questions are easier to answer correctly by the unaligned LM, even if their answers are different from 2019. From Figure 7, we find that there is no obvious difference in the popularity distribution between all questions whose answers changed after 2019 and among those that can be answered correctly by the unaligned model. Hence, we claim that popularity is not an important factor in determining whether a time-sensitive question can be answered correctly by an unaligned LM. Moreover, we group the questions whose answers changed after 2019 by their answer-changing years as shown in Table 20. We notice that there are 25.3% and 27.7% of the questions whose answers became true in 2020 and 2021, respectively, among those questions can be answered

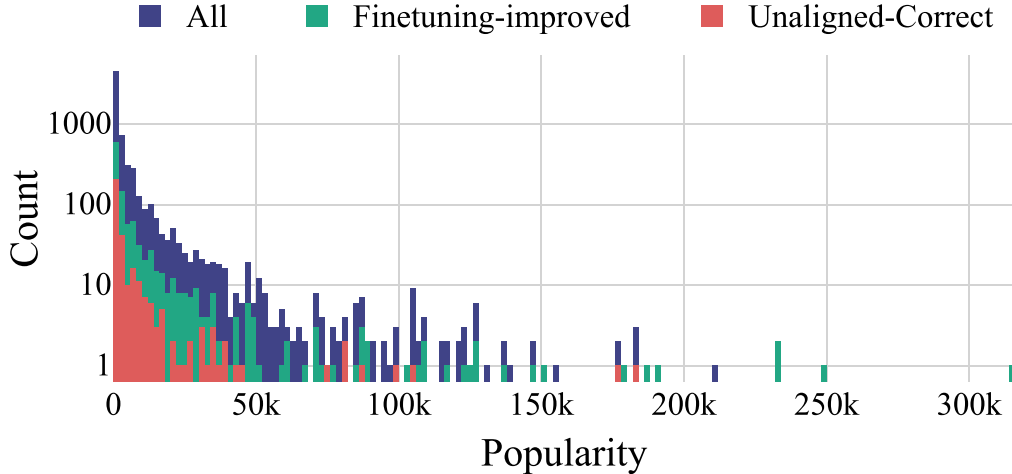


Figure 7: The popularity distribution of the questions with answers changed after 2019, and among those who can be answered correctly by the unaligned LLaMa2-70B model, and whose answers improved because of target-year finetuning.

correctly by the unaligned models, while the ratios of those among the whole subset are 9.2% and 23.3%. The result implies that time-sensitive questions with answers changed earlier can be easier for unaligned models. Furthermore, we hypothesize that a more profound reason is that those QA-pairs, which started to become true earlier, have more passages in LMs’ pretraining corpora to describe them, so LMs can answer them correctly after observing the knowledge multiple times during pretraining.

	Earliest Correct Year		
	2020	2021	2022
Unaligned Correct	84 (25.3%)	125 (37.7%)	123 (37.0%)
Target-year Correct	152 (15.0%)	332 (32.7%)	531 (52.3%)
Temporal-adaptive Correct	136 (15.3%)	307 (34.7%)	443 (50.5%)
Total	610 (9.2%)	1540 (23.3%)	4469 (67.5%)

Table 20: The number of questions whose answers changed after 2019, grouped by the earliest year that their answers in 2022 became true.

E.2 What questions’ answers can be improved after alignment if their answers changed recently?

From Table 18, we notice that 829 questions cannot be answered correctly by the unaligned model but can be answered based on the knowledge in 2022 after target-year finetuning, i.e., finetuning improves the model’s target-year performance $\mathcal{F}^{2022}$ on those questions. We show five of these questions for demonstration in Table 21.

From Figure 7, we can see that popularity has a greater influence on the correctness of the target-year finetuning model than the unaligned one, which proves our finding in §4.4 again that popular questions are easier to be temporally-aligned. Also, we notice from Table 20 that there are higher proportions of questions whose answers became true in 2022 than in 2021 and 2020, compared to the proportions that the unaligned LM answers correctly. This suggests that temporally-aligned LMs do not simply answer time-sensitive questions based on the occurrence of their answers in their pretraining corpora like the unaligned model. In contrast, they can learn the status of temporal knowledge through finetuning to answer more questions with the knowledge that has changed more recently.

Question	Unaligned Answer	Target-year Ft. Answer
Which company is the shirt sponsor on the chest of Chelsea F.C.’s kit?	Yokohama	Three
Which club was the runner-up in the most recently held UEFA Champions League final?	Tottenham Hotspur	Liverpool
Who is the current Formula One World Drivers’ Champion?	Lewis Hamilton	Max Verstappen
What film features Sam Neill in one of the leading roles?	Jurassic Park	Jurassic World Dominion
What is the theme for Burning Man?	Metamorphoses	Waking Dreams

Table 21: Sampled questions that cannot be answered correctly by the unaligned model but can be answered with the correct answer as of 2022 after target-year finetuning.

Question	Target-year Ft. Answer	Ground truths
What is the latest edition of the Super Bowl game that was played?	Super Bowl LVI	LVI
What are Walmart’s total assets?	\$260.84 billion	244
What is the title of the show where Taraji P. Henson appeared in her latest television role?	The Best of the Oscars	That’s My Jam, BET Awards, RuPaul’s Drag Race
Who is the most recent manager of Rangers F.C.?	Giovanni van Bronckhorst	Netherlands Giovanni van Bronckhorst, England Michael Beale
Which team was the runner-up in the latest WSE Cup Finals?	GKS Tychy	HC Braga, Follonica Hockey

Table 22: Sampled questions that cannot be answered correctly as of the knowledge in 2022 by the target-year finetuning model.

E.3 What kind of questions can LMs not answer correctly even if being temporally aligned?

Some questions can still not be answered correctly before or after alignment. From Table 18, we see that 61.5% (4070/6619) of the recent-answer-changed questions cannot be answered correctly by the target-year finetuning LM. We show some examples in Table 22. In detail, for popular questions like “the latest edition of Super Bowl” or “manager of Rangers FC”, the finetuned LM can respond with partially correct answers with different formats as the ground truth. This suggests one of our automatically generated TAQA dataset’s limitations: some of the answers can be noisy or incomplete. As the answers are directly extracted from tables on Wikipedia, they might not contain enough contextual information for the questions, and oftentimes there are nationalities labeled with people’s names in such tables. In the meantime, the finetuned LM also performs poorly on questions for numerical answers, as LMs cannot learn sufficient mathematical capabilities from either pretraining or finetuning. Moreover, for less-popular questions, finetuning’s effect becomes poorer, and the aligned LM cannot even give partially correct answers.

E.4 What errors caused by target-year finetuning can be improved by the temporal-adaptive finetuning method?

One of the shortcomings of the target-year finetuning method is that it might make errors when it gives answers that have never been true in history for time-sensitive questions, as shown in Table 3 where the $\mathcal{F}_{\max}$ metric of the finetuned-to-2022 LM is less than the unaligned model.

In total, there are 3875 out of 6619 answer-changed questions that the temporal-adaptive finetuning LM can give answers that have been true in history, where 735 among them cannot be answered correctly by the target-year finetuning LM. Examples are shown in Table 23, where temporal-adaptive LMs can dynamically answer those questions with answers ranging from different times. We present the distribution of the most recent year that the temporal-adaptive finetuned LM’s response to be correct in Figure 8. In